Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/15/2024
PI Name:	Thoolen, Stijn MD		
Project Title:	Comparison of the Negative Effects of Carbon Dioxide Exposure Among Ground, Hypobaric, and Spaceflight Environments (PI: Thoolen)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **EVA**:Risk of Mission Impacting Injury and Compromised Performance and Long-Term Health Effects due to EVA Operations		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	
Organization Name:	KBR/NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/04/2022	**End Date:**	04/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Dr. Nusbaum left the project as the study Principal Investigator (PI) in October 2023; Dr. Stijn Thoolen has been assigned as the new study PI. Dr. Abercromby, Dr. Dunn, and Alex Baughman were removed as Co-Investigators, as they have left the project. Dr. Marshall-Goebel, Dr. Garbino, Dr. Basner, and Dr. Bell were added as Co-Investigators.		
COI Name (Institution):	Marshall-Goebel, Karina Ph.D. (NASA Johnson Space Center) Garbino, Alejandro M.D., Ph.D. (KBR/NASA Johnson Space Center) Scully, Robert Ph.D. (NASA Johnson Space Center) Levine, Benjamin M.D. (University of Texas Southwestern Medical Center) Babb, Tony Ph.D. (University of Texas Southwestern Medical Center) Pawelczyk, James Ph.D. (Pennsylvania State University) Basner, Mathias M.D., Ph.D. (University of Pennsylvania) Bell, Suzanne Ph.D. (NASA Johnson Space Center) Young, Millennia (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Carbon dioxide (CO2) produced through the metabolism of astronauts inside spacesuits may accumulate during extravehicular activities (EVA). The maximum level of inspired CO2 during a 1-hour contingency walkback scenario is unknown. The CO2 Walk Back Study (See “Effects of Carbon Dioxide Exposure on Physical and Cognitive Performance in a Simulated Spaceflight Contingency Scenario”, PI Garbino), which is a companion study to this study, will characterize the limit of acceptable performance (cognitive and physical) decrements and symptom severity during a simulated 1-hour contingency walk back when individuals are subjected to elevated inspired CO2 levels (up to 30 mmHg) in the laboratory. However, there are concerns that the lower pressures, higher oxygen (O2) concentrations, and/or altered gravity environment that exist in the exploration spaceflight scenario may significantly impact the response to CO2. The purpose of this study is to quantify the extent to which responses to CO2 are affected by the spacesuit atmosphere (reduced pressure with slightly elevated oxygen levels) and the spaceflight (altered gravity) environment. The high-fidelity EVA contingency walk back simulation task (which includes Virtual Reality and a treadmill) is not feasible to implement in the hypobaric and spaceflight environments, so this study will use an alternative EVA simulation task while recording the same physical and cognitive performance measures as the CO2 Walk Back Study. Results from this study will be used to either validate that CO2 dose-response relationships from the CO2 Walk Back Study are directly applicable to contingency EVA scenarios during exploration spaceflight, or else provide a recommended quantitative adjustment to dose-response relationships to account for the effects of the spacesuit atmosphere and/or gravity. Together, these studies will help to determine the maximum allowable inspired CO2 level that will still enable crewmembers to complete a one-hour contingency EVA walk back within acceptable functional performance limits. Furthermore, as this study will include CO2 response data at nominal partial pressures of inspired CO2 in addition to contingency levels, this study will also inform and clarify overall risk of CO2 exposure in the flight environment – not just for suited EVA operations – which has been deemed an important human health and performance risk for future spaceflight missions. Methods: This study will use two experimental arms – a hypobaric chamber arm (n=16) and a flight arm (n=5). For each arm, we will use a repeated measures within-subject single-blinded design. In both the Chamber and Flight Arms, subjects will serve as their own control by undergoing the same experiments in a ground-based laboratory environment. The laboratory experiments will be performed in Johnson Space Center’s Building 21. The chamber experiments will be performed in the 20-ft Chamber in NASA Johnson Space Center’s Building 7. The flight experiments will be performed on the ISS. All experiments will be performed in a shirtsleeve configuration, using a cycle ergometer to simulate the metabolic rates of a contingency walk back scenario during a lunar surface EVA. During each experiment, subjects will repeatedly perform a standardized one-hour exercise and performance assessment protocol while being exposed to a breathing gas with varying partial pressures of inspired CO2. Physical performance will be measured by evaluating the duration subjects are able to maintain the target workload (up to a maximum of one hour) and quantifying the physiological (cardiopulmonary) effort to do the work. Cognitive performance will be quantified using the Cognition cognitive test battery (Basner et al., 2015) and repeated Digital Symbol Substitution Tasks (DSST). The severity of specific CO2-related symptoms and a subjective self-assessment of performance will be quantified via survey. References: Norcross J, Abercromby AF, Alexander D. Maximum xEMU CO2 During Emergency Abort. 2020. Joint BRESCB/SMOCB Forum. Basner M, Savitt A, Moore TM, Port AM, McGuire S, Ecker AJ, Nasrini J, Mollicone DJ, Mott CM, McCann T, Dinges DF, Gur RC. Development and Validation of the Cognition Test Battery for Spaceflight. Aerosp Med Hum Perform. 2015 Nov;86(11):942-52.
Rationale for HRP Directed Research:	This project is directed because it contains highly constrained research. This is a rapid response study of operational relevance, informing future standards and requirements and thus driving future hardware design for future programs. This project is a companion to its counterpart “Effect of Carbon Dioxide Exposure on Physical and Cognitive Performance in a Simulated Spaceflight Contingency Scenario” and is necessary to extrapolate the results of the Rapid Response ground study to the altered gravity and atmospheric environments expected during exploration missions. It should therefore be directed to the same PI team.
Research Impact/Earth Benefits:	The principal benefits of this study are expected to be in the area of operational space medicine, where these data will help inform future planetary EVA suit design and may drive EVA operational constraints. The results may also be used to drive habitat design and operations. However, as this study deals with the cognitive and physical performance aspects of elevated CO2 exposure, it may have broader implications for clinical patients who are also exposed to elevated CO2, such as patients with chronic obstructive lung disease or patients with obesity hypoventilation syndrome.
Task Progress:	Ed. Note: Project has closed. Per NASA Johnson Space Center (JSC) no final performance report or data deliverables forthcoming (Ed., 4/30/25). ---------- Continuation of "Comparison of the Negative Effects of Carbon Dioxide Exposure Among Ground, Hypobaric, and Spaceflight Environments". See project (with same title) under Dr. Derek Nusbaum for earlier reporting (Ed., 8/15/24). In 2023/2024, the study protocol and design was revisited based on feasibility/implementation constraints. A proof-of-concept prototype of a CO2 rebreather for use in the International Space Station (ISS) arm of the study was finished in June 2024 and the methodology is currently undergoing a feasibility analysis by NASA's Research Operations and Integration (ROI) Element for inflight implementation. In addition, preparation of a CO2 delivery system to be used in the hypobaric chamber was developed and implemented. A 1-day equipment acceptance test inside the chamber has been scheduled in September 2024, to test the safety, proper functioning, and feasibility of the hardware before final implementation in the chamber. The first 9-day test run in the hypobaric chamber is currently

	scheduled for September 2025. The second 9-day chamber run is provisionally scheduled for November 2025. Laboratory-based testing for the Chamber Arm in Building 21 has currently been scheduled for June 2025.
Bibliography Type:	Description: (Last Updated:)